

DEPARTMENT OF INFORMATION TECHNOLOGY

24ITT32 - PROGRAMMING IN JAVA

INDIVIDUAL PROJECT ON
ACADEMIC MANAGEMENT SYSTEM

PROJECT REPORT

Submitted by

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JAVA ASSIGNMENT

AIM:

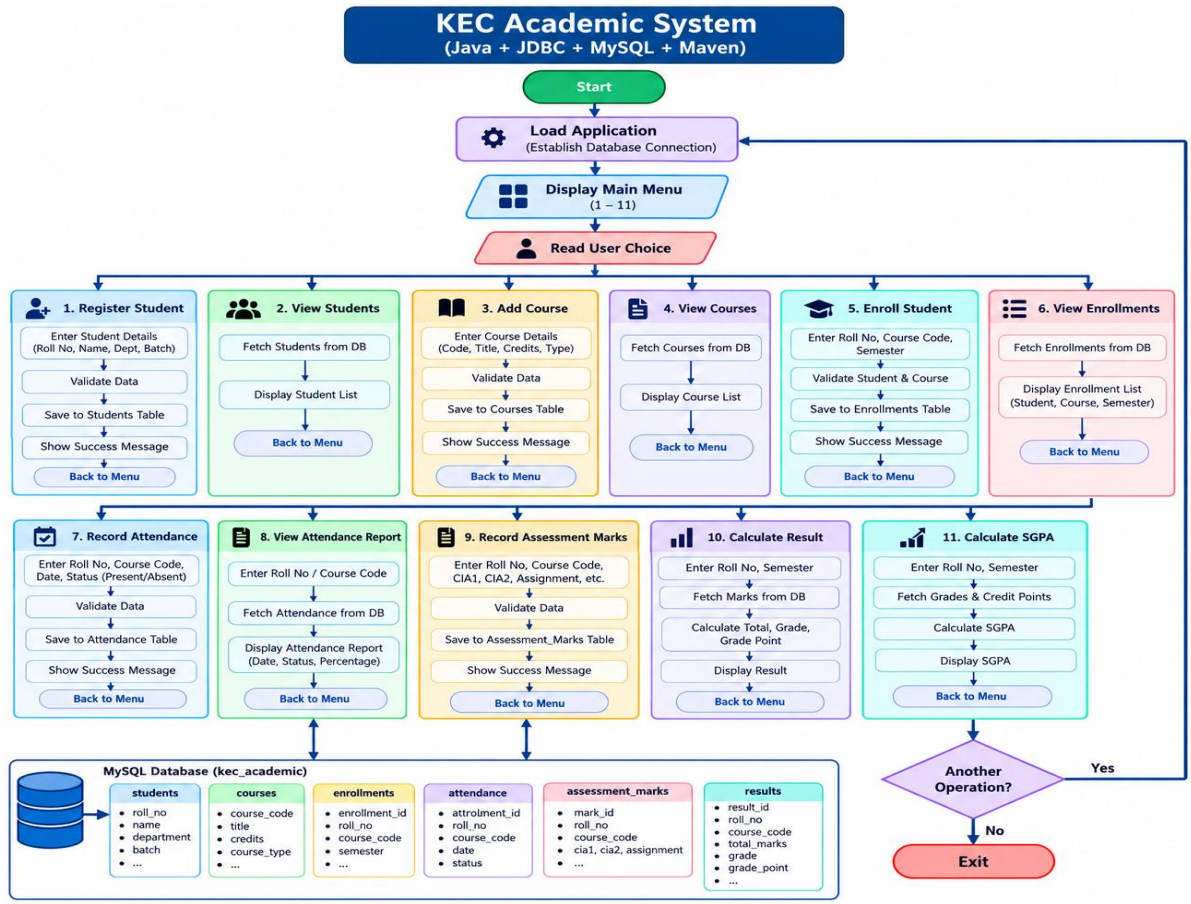
Development of Kongu Engineering College (R2024) Student Academic & Continuous Evaluation System using Java JDBC.

OBJECTIVE:

Design and develop a database-driven Java application using JDBC to manage student records, perform incremental marks entry (Continuous Internal Assessments and End Semester Examinations), enforce attendance eligibility rules, and generate formal academic performance reports according to

Kongu Engineering College (Autonomous) - Regulation 2024. System Requirements & Specifications.

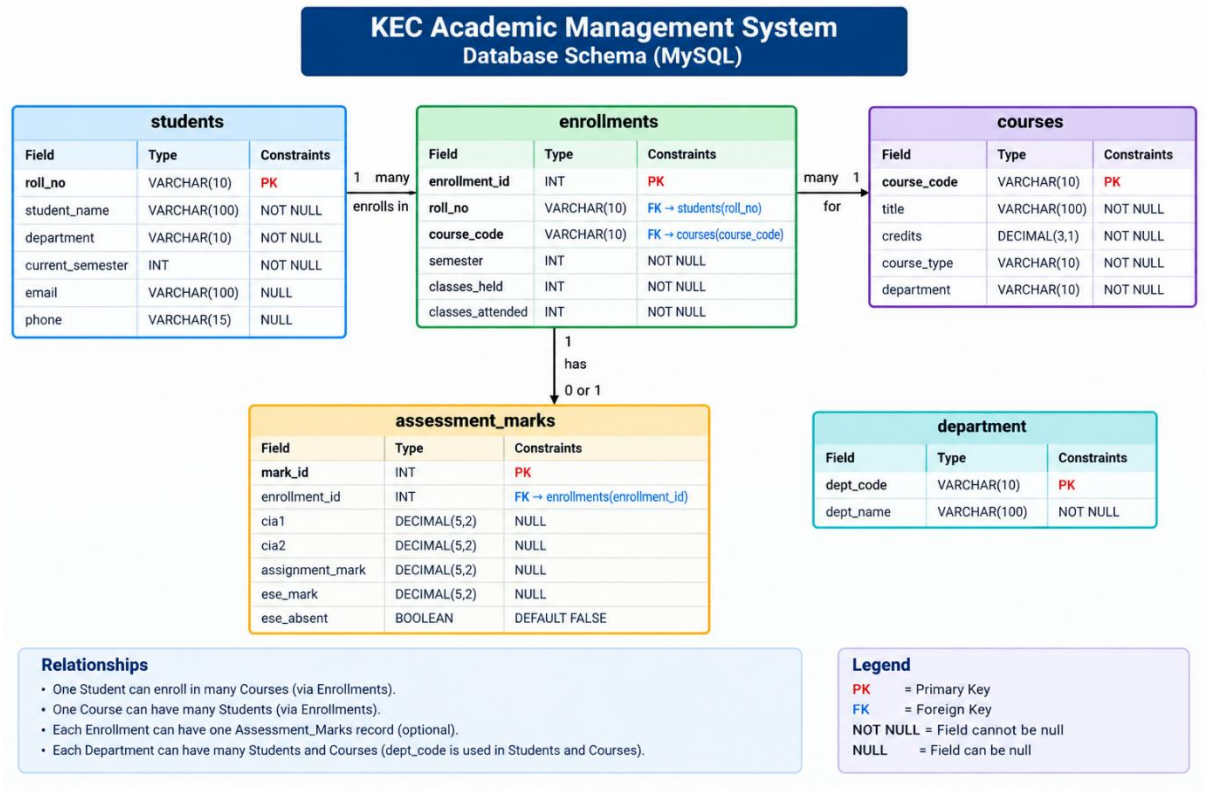
FLOW CHART:



PROCEDURE:

1. Install and configure Java JDK, VS Code, MySQL Server, and MySQL Connector/J.
2. Create a Java project named KEC-Academic-System and add the MySQL Connector/J library.
3. Create the database kec_academics in MySQL and define the required tables for students, courses, enrollments, and assessment marks.
4. Create DBConnection.java to establish a connection between Java and MySQL using JDBC.
5. Create StudentDAO.java to register and display student details.
6. Create AcademicDAO.java to manage courses, enrollment, attendance, marks, grades, and academic reports.
7. Create MainApp.java to display the menu and call the appropriate methods based on the user's choice.
8. Execute the SQL script and verify the database connection.
9. Run MainApp.java and test student registration, course enrollment, attendance updates, and marks entry.
10. Verify the stored records and academic results to ensure the system functions correctly.

DATABASE SCHEMA:



SOURCE CODE:

1. DBConnection.java

```
package kec;


import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;


public class DBConnection {
    private static final String URL =
            System.getenv().getOrDefault("KEC_DB_URL",
"jdbc:mysql://localhost:3306/kec_academic");
    private static final String USER = System.getenv().getOrDefault("KEC_DB_USER", "root");
    public static Connection getConnection() throws SQLException {
        String password = System.getenv("KEC_DB_PASSWORD");
        if (password == null || password.isBlank()) {
            throw new SQLException("KEC_DB_PASSWORD environment variable is required.");
        }
        return DriverManager.getConnection(URL, USER, password);
    }
    public static void main(String[] args) {
        try (Connection connection = getConnection()) {
            System.out.println("MySQL database connected successfully!");
        } catch (SQLException e) {
            System.out.println("Database connection failed!");
            e.printStackTrace();
        }
    }
}
```

2. StudentDAO.java

```
package kec;

import java.sql.Connection;
import java.sql.PreparedStatement;
import java.sql.SQLException;

public class StudentDAO {
    public boolean addStudent(String rollNo, String name,
                              String department, int semester) {
        String sql = "INSERT INTO students " +
                     "(roll_no, student_name, department, current_semester) " +
                     "VALUES (?, ?, ?, ?)";
        try (Connection con = DBConnection.getConnection();
             PreparedStatement ps = con.prepareStatement(sql)) {

            ps.setString(1, rollNo);
            ps.setString(2, name);
            ps.setString(3, department);
            ps.setInt(4, semester);

            int rows = ps.executeUpdate();
            if (rows > 0) {
                System.out.println("Student registered successfully!");
                return true;
            }
        } catch (SQLException e) {
            System.out.println("Error registering student: " + e.getMessage());
        }
        return false;
    }
}
```

3.MainApp.java

```
package kec;

import java.util.Scanner;
public class App {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        StudentDAO studentDAO = new StudentDAO();
        AcademicDAO academicDAO = new AcademicDAO();
        CourseDAO courseDAO = new CourseDAO();
        EnrollmentDAO enrollmentDAO = new EnrollmentDAO();
        AttendanceDAO attendanceDAO = new AttendanceDAO();
        AssessmentDAO assessmentDAO = new AssessmentDAO();
        ResultDAO resultDAO = new ResultDAO();
        SGPADao sgpaDAO = new SGPADao();

        while (true) {

            System.out.println("\n==================================");
            System.out.println("        KEC ACADEMIC SYSTEM");
            System.out.println("==================================");
            System.out.println("1. Register Student");
            System.out.println("2. View Registered Students");
            System.out.println("3. Add Course");
            System.out.println("4. View Courses");
            System.out.println("5. Enroll Student in Course");
            System.out.println("6. View Enrollments");
            System.out.println("7. Record Attendance");
            System.out.println("8. View Attendance Report");
            System.out.println("9. Record Assessment Marks");
```

```
System.out.println("10. Calculate Student Result");
System.out.println("11. Calculate SGPA");
System.out.println("12. Exit");
System.out.println("====================================");
System.out.print("Enter your choice (1-12): ");

if (!sc.hasNextInt()) {
    System.out.println("Invalid input! Enter a number.");
    sc.nextLine();
    continue;
}

int choice = sc.nextInt();
sc.nextLine();

switch (choice) {

    case 1:
        System.out.print("Enter Roll Number: ");
        String rollNo = sc.nextLine();

        System.out.print("Enter Student Name: ");
        String studentName = sc.nextLine();
        System.out.print("Enter Department: ");
        String department = sc.nextLine();
        System.out.print("Enter Current Semester: ");
        if (!sc.hasNextInt()) {
            System.out.println("Invalid semester!");
            sc.nextLine();
            break;
        }
        int semester = sc.nextInt();
```

```
            sc.nextLine();
            studentDAO.addStudent(
                rollNo, studentName, department, semester
            );
            break;
        case 2:
            academicDAO.viewStudents();
            break;
        case 3:
            System.out.print("Enter Course Code: ");
            String courseCode = sc.nextLine();

            System.out.print("Enter Course Title: ");
            String courseTitle = sc.nextLine();

            System.out.print("Enter Credits: ");
            if (!sc.hasNextInt()) {
                System.out.println("Invalid credits!");
                sc.nextLine();
                break;
            }
            int credits = sc.nextInt();
            sc.nextLine();

            System.out.print(
                "Enter Course Type (THEORY/PRACTICAL/EMBEDDED): "
            );
            String courseType = sc.nextLine().trim().toUpperCase();

            courseDAO.addCourse(
                courseCode, courseTitle, credits, courseType
            );
```

```
            break;
        case 4:
            courseDAO.viewCourses();
            break;
        case 5:
            System.out.print("Enter Student Roll Number: ");
            String enrollRollNo = sc.nextLine();

            System.out.print("Enter Course Code: ");
            String enrollCourseCode = sc.nextLine();

            System.out.print("Enter Semester: ");
            if (!sc.hasNextInt()) {
                System.out.println("Invalid semester!");
                sc.nextLine();
                break;
            }
            int enrollSemester = sc.nextInt();
            sc.nextLine();

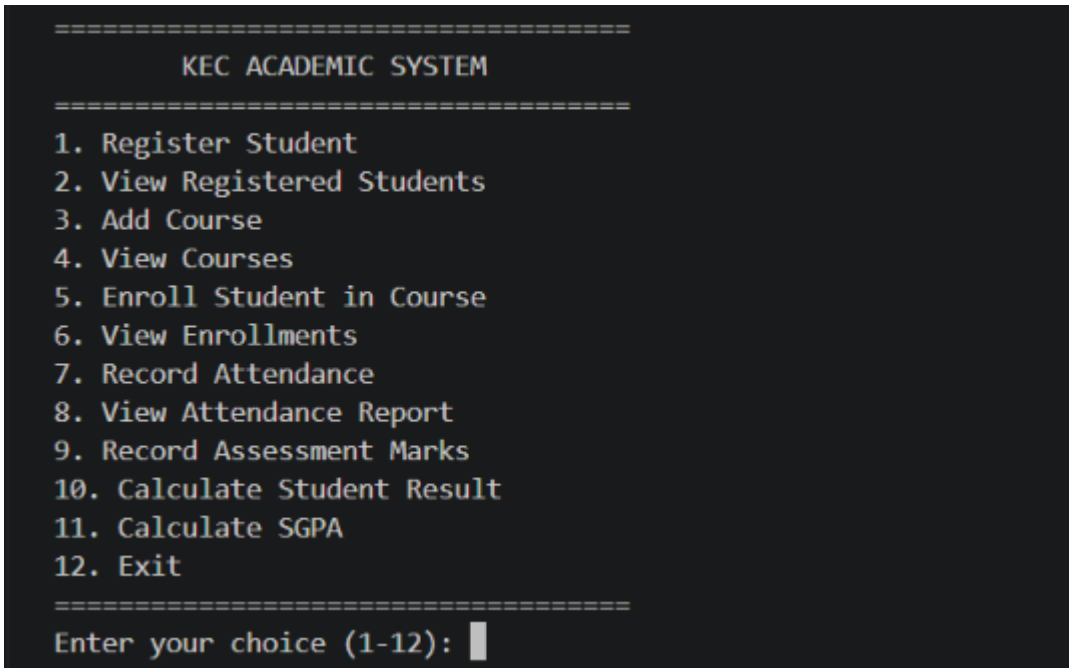
            enrollmentDAO.enrollStudent(
                enrollRollNo, enrollCourseCode, enrollSemester
            );
            break;
        case 6:
            enrollmentDAO.viewEnrollments();
            break;
        case 7:
            System.out.print("Enter Student Roll Number: ");
            String attendanceRollNo = sc.nextLine();
            System.out.print("Enter Course Code: ");
            String attendanceCourseCode = sc.nextLine();
```

```
System.out.print("Enter Semester: ");
if (!sc.hasNextInt()) {
    System.out.println("Invalid semester!");
    sc.nextLine();
    break;
}
int attendanceSemester = sc.nextInt();
System.out.print("Enter Classes Held: ");
if (!sc.hasNextInt()) {
    System.out.println("Invalid number of classes!");
    sc.nextLine();
    break;
}
int classesHeld = sc.nextInt();
System.out.print("Enter Classes Attended: ");
if (!sc.hasNextInt()) {
    System.out.println("Invalid number of classes!");
    sc.nextLine();
    break;
}
int classesAttended = sc.nextInt();
sc.nextLine();
System.out.print(
    "Medical Condonation Approved? (Y/N): "
);
String medicalInput = sc.nextLine().trim();
boolean medicalCondonation =
    medicalInput.equalsIgnoreCase("Y");
attendanceDAO.recordAttendance(
    attendanceRollNo,
    attendanceCourseCode,
    attendanceSemester,
```

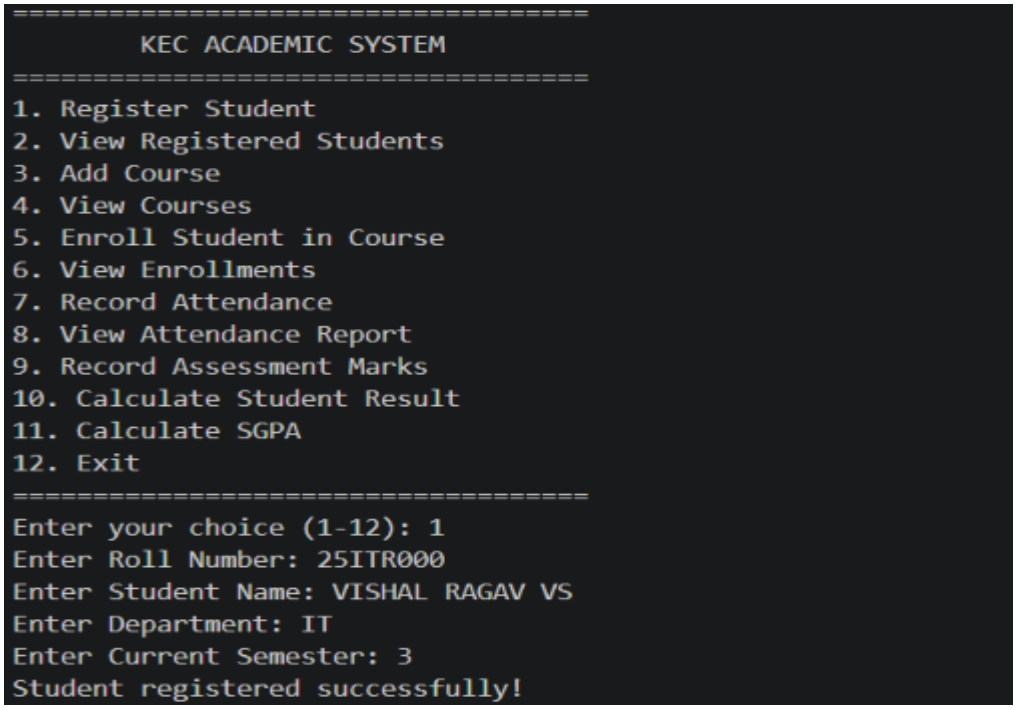
```
                    classesHeld,
                    classesAttended,
                    medicalCondonation
                );
                break;
            case 8:
                attendanceDAO.viewAllAttendance();
                break;
            case 9:
                assessmentDAO.recordAssessmentMarks();
                break;
            case 10:
                resultDAO.calculateStudentResult();
                break;
            case 11:
                sgpaDAO.calculateSGPA();
                break;
            case 12:
                System.out.println(
                    "Thank you for using KEC Academic System!"
                );
                sc.close();
                return;
            default:
                System.out.println(
                    "Invalid choice! Enter a number from 1 to 12."
                );
        }
    }
  }
}
```

OUTPUT SCREENSHOTS:

1.MAIN MENU:



2.STUDENT REGISTRATION:



3. COURSE REGISTRATION:

```
==================================
        KEC ACADEMIC SYSTEM
==================================
1. Register Student
2. View Registered Students
3. Add Course
4. View Courses
5. Enroll Student in Course
6. View Enrollments
7. Record Attendance
8. View Attendance Report
9. Record Assessment Marks
10. Calculate Student Result
11. Calculate SGPA
12. Exit
==================================
Enter your choice (1-12): 3
Enter Course Code: 24ITT32
Enter Course Title: PROGRAMMING IN JAVA
Enter Credits: 4
Enter Course Type (THEORY/PRACTICAL/EMBEDDED): THEORY
Course added successfully!
```

4.STUDENT ENROLLMENT :

```
==================================
        KEC ACADEMIC SYSTEM
==================================
1. Register Student
2. View Registered Students
3. Add Course
4. View Courses
5. Enroll Student in Course
6. View Enrollments
7. Record Attendance
8. View Attendance Report
9. Record Assessment Marks
10. Calculate Student Result
11. Calculate SGPA
12. Exit
==================================
Enter your choice (1-12): 5
Enter Student Roll Number: 25ITR000
Enter Course Code: 24ITT32
Enter Semester: 3
Student enrolled successfully!
```

5. MARKS ENTRY:

```
==================================
        KEC ACADEMIC SYSTEM
==================================
1. Register Student
2. View Registered Students
3. Add Course
4. View Courses
5. Enroll Student in Course
6. View Enrollments
7. Record Attendance
8. View Attendance Report
9. Record Assessment Marks
10. Calculate Student Result
11. Calculate SGPA
12. Exit
==================================
Enter your choice (1-12): 9

===== RECORD ASSESSMENT MARKS =====
Enter Student Roll Number: 25ITR000
Enter Course Code: 24ITT32
Enter Semester: 3
Enter CIA1 marks (0-100): 88
Enter CIA2 marks (0-100): 88
Enter Assignment marks (0-100): 90
Enter Quiz marks (0-100): 90
Enter Lab Evaluation marks (0-100, enter 0 if not applicable): 90
Was the student absent for ESE? (Y/N): N
Enter ESE marks (0-60): 55

Assessment marks saved successfully!
```

6.ATTENDENCE ENTRY:

```
==================================
        KEC ACADEMIC SYSTEM
==================================
1. Register Student
2. View Registered Students
3. Add Course
4. View Courses
5. Enroll Student in Course
6. View Enrollments
7. Record Attendance
8. View Attendance Report
9. Record Assessment Marks
10. Calculate Student Result
11. Calculate SGPA
12. Exit
==================================
Enter your choice (1-12): 7
Enter Student Roll Number: 25ITR000
Enter Course Code: 24ITT32
Enter Semester: 3
Enter Classes Held: 60
Enter Classes Attended: 55
Medical Condonation Approved? (Y/N): N
Attendance recorded successfully!
Attendance: 91.67% (55/60 classes)
Status: ESE Eligible
```

7.DISPLAY ATTENDANCE:

```
==================================
        KEC ACADEMIC SYSTEM
==================================
1. Register Student
2. View Registered Students
3. Add Course
4. View Courses
5. Enroll Student in Course
6. View Enrollments
7. Record Attendance
8. View Attendance Report
9. Record Assessment Marks
10. Calculate Student Result
11. Calculate SGPA
12. Exit
==================================
Enter your choice (1-12): 8

===== ATTENDANCE REPORT =====

Roll Number: 25ITR000
Student: VISHAL RAGAV VS
Course: 24ITT32 - PROGRAMMING IN JAVA
Semester: 3
Attendance: 91.67% (55/60)
Status: ESE Eligible
```

8.STUDENT RESULT:

```
===== STUDENT RESULT CALCULATION =====
Enter Student Roll Number: 25ITR000
Enter Course Code: 24ITT32
Enter Semester: 3

========================================
          STUDENT RESULT
========================================
Roll Number : 25ITR000
Name        : VISHAL RAGAV VS
Department  : IT
Course Code : 24ITT32
Course Name : PROGRAMMING IN JAVA
Course Type : THEORY
Credits     : 4
Attendance  : 91.67%
Attendance Status: ESE Eligible
CIA Average : 89.20 / 100
CIA Weighted: 35.68
ESE Marks   : 55.00 / 60
ESE Weighted: 55.00
Final Total : 90.68 / 100
Grade       : A+
Grade Point : 9
Status      : PASS
```

9.STUDENT SGPA:

```
===== SGPA CALCULATION =====
Enter Student Roll Number: 25ITR000
Enter Semester: 3

================================================================
                  SEMESTER GRADE REPORT
================================================================
Code       Course                    Credits    Grade   Points
----------------------------------------------------------------
24ITT32    PROGRAMMING IN JAVA             4       A+        9
----------------------------------------------------------------
Student Name: VISHAL RAGAV VS
Department  : IT
Semester    : 3
Courses     : 1
Total Credits: 4
SGPA        : 9.00 / 10.00
Percentage  : 90.00%
================================================================
```

RESULT:

Thus, the Student Academic and Continuous Evaluation System for Kongu Engineering College (Autonomous) Regulation 2024 was successfully designed, developed, and tested using Java (JDBC) and MySQL. The system accurately automates the calculation of Continuous Internal Assessments (CIA), End Semester Examinations (ESE), Letter Grades, Attendance Eligibility, and Semester Grade Point Average (SGPA).

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GITHUB:

LINK:	https://github.com/VISHAL51053/java_JDBC